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Laboratory Report – 8

Objective

Generating DSM, DEM, Slope map, Aspect map, Channel Network for a given area using LiDAR data

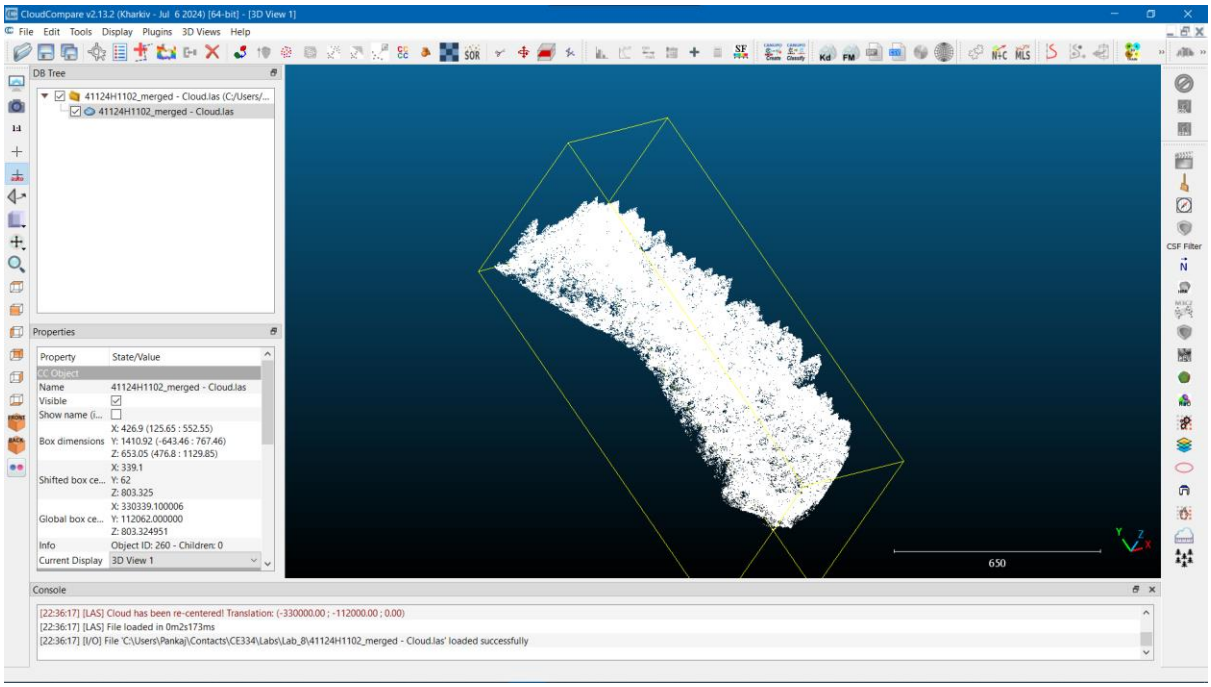
Report

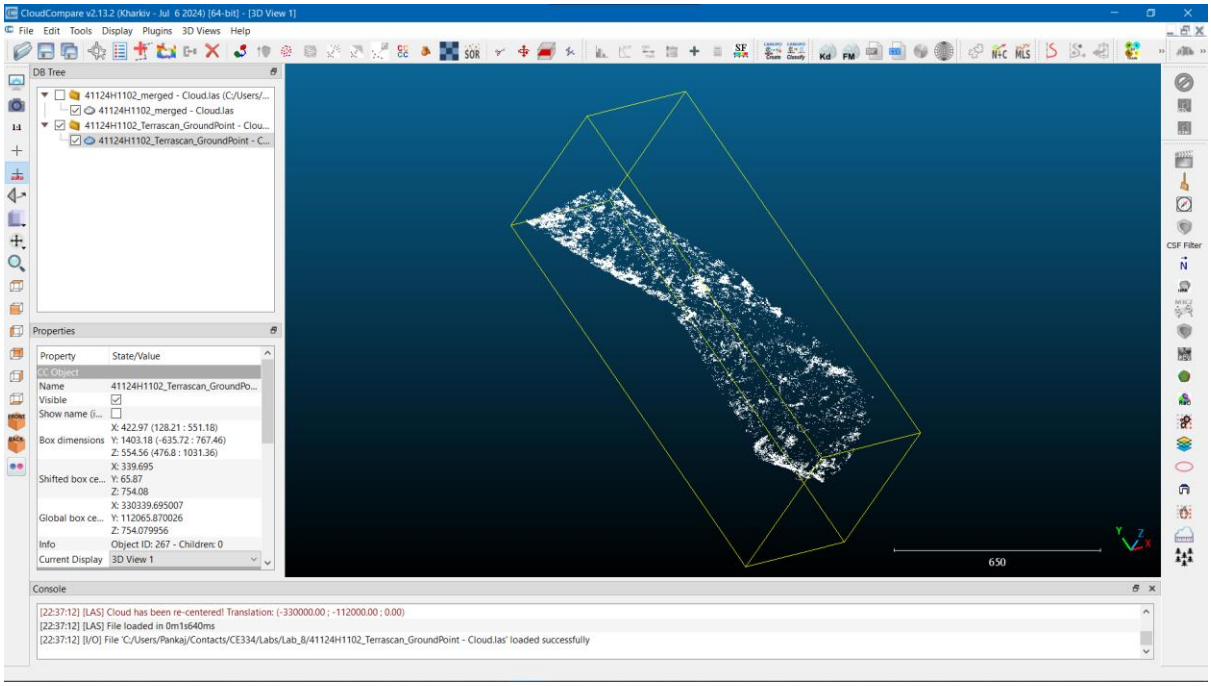
Report should contain the following:

- Snapshots of original data display.
- Snapshots of DSM, DEM, Slope Map, Aspect Map, Channel Network.
- Write your thoughts about all generated maps in the report.
- Write answer to the questions asked above regarding applicability of thresholding method for ground and non-ground classification.
- Write one use of each of data derivative generated above (Write answers in very brief and to the point.)

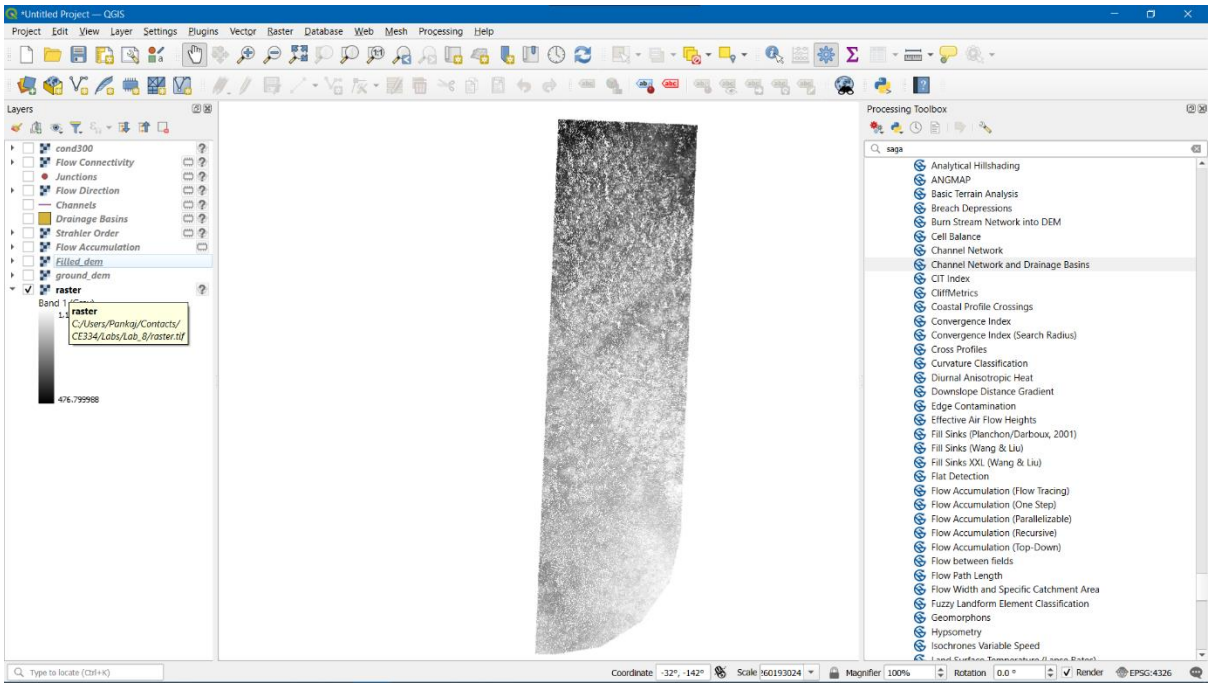
Methodology with Q-GIS and CloudCompare

- The file used in this lab are i.e. 41124H1102_merged – Cloud and 41124H1102_Terrascan_GroundPoint – Cloud.

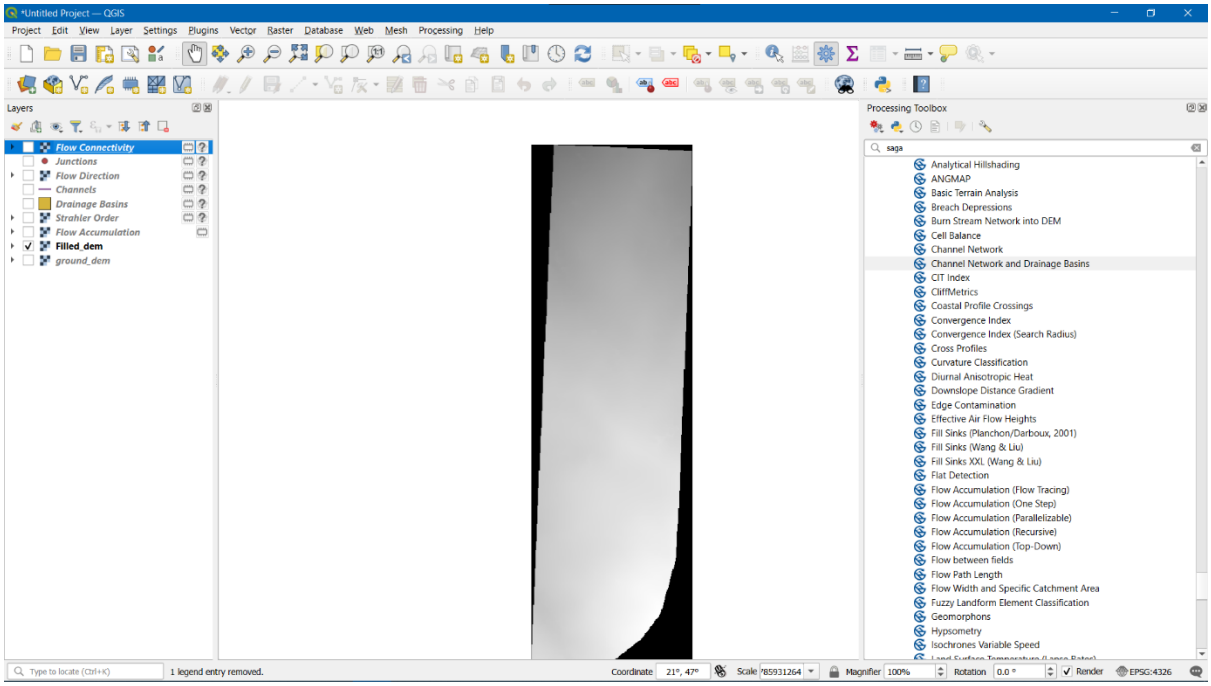




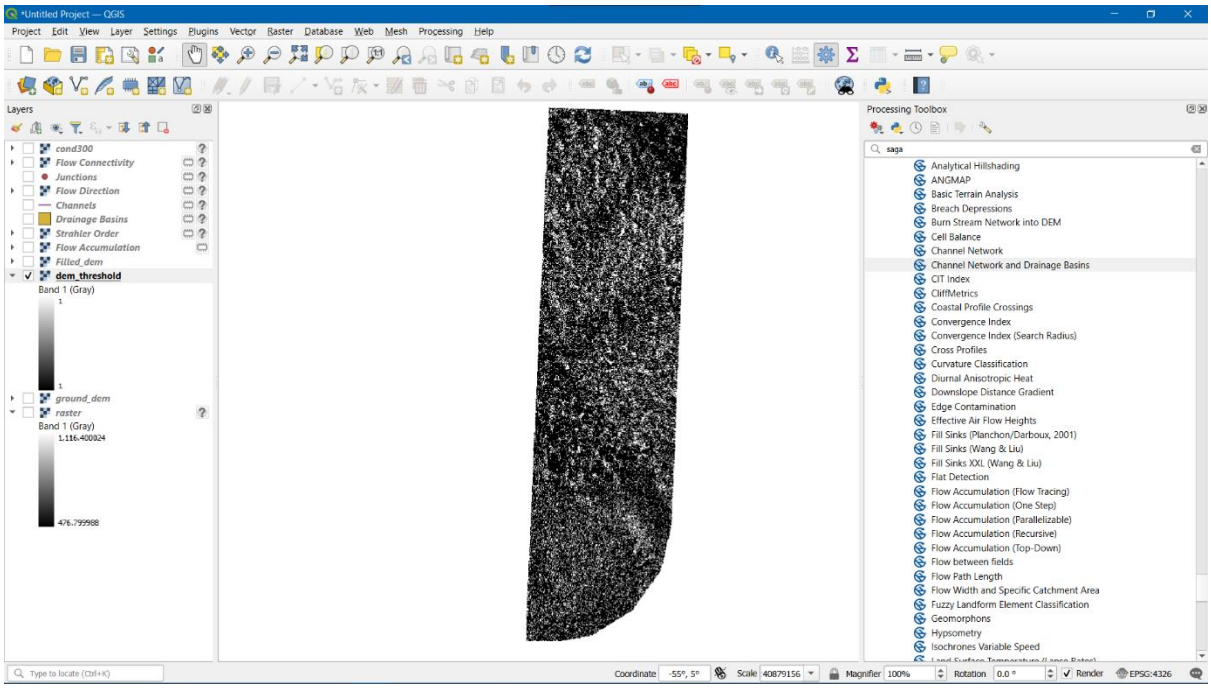
- Observe the point data of both files in a CloudCompare.
- Generate DSM



- Choose appropriate size of raster and cell size and generate DSM using Raw data file.
- Choose same sizes as above and generate DEM using Ground Points.



➢ Try to generate the DEM using the thresholding technique



➢ Compare the output generated by thresholding with the ground class data provided.

High Threshold: Results in a sparse, less detailed channel network representing only major rivers and streams.

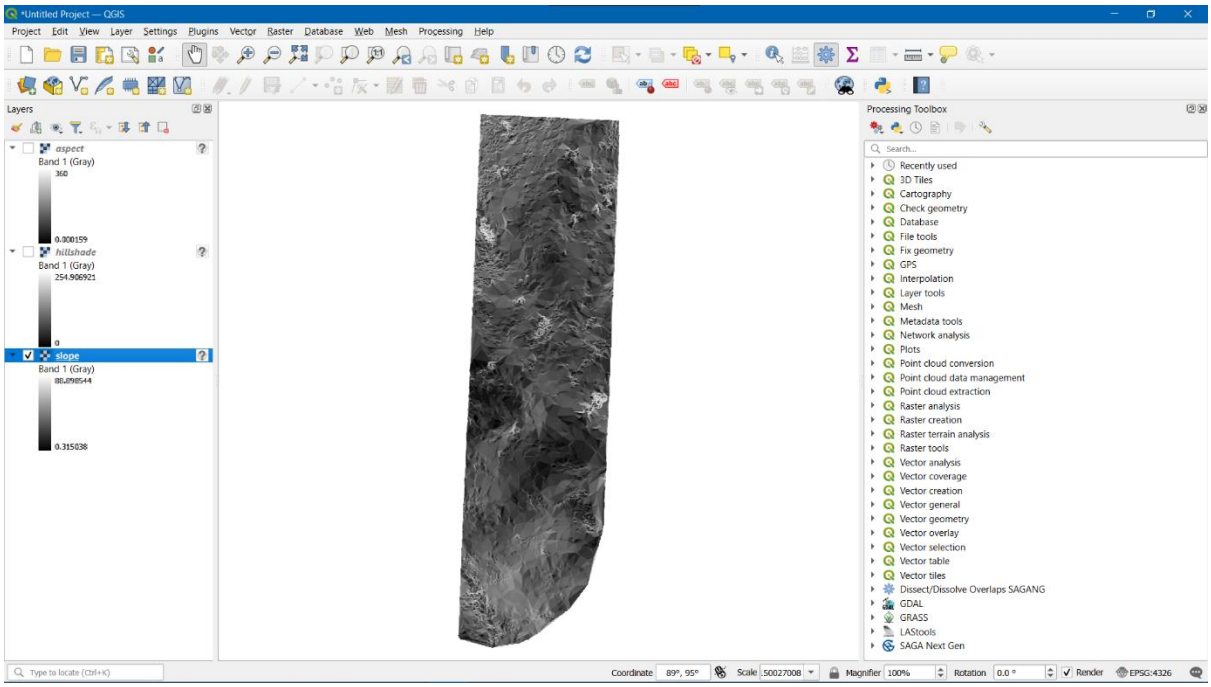
Low Threshold: Results in a dense, highly branched network that includes minor rills, gullies, and small tributaries.

➢ ***Question*: Can we classify data in ground and above-ground points using thresholding? If yes, then how? If no, then why?**

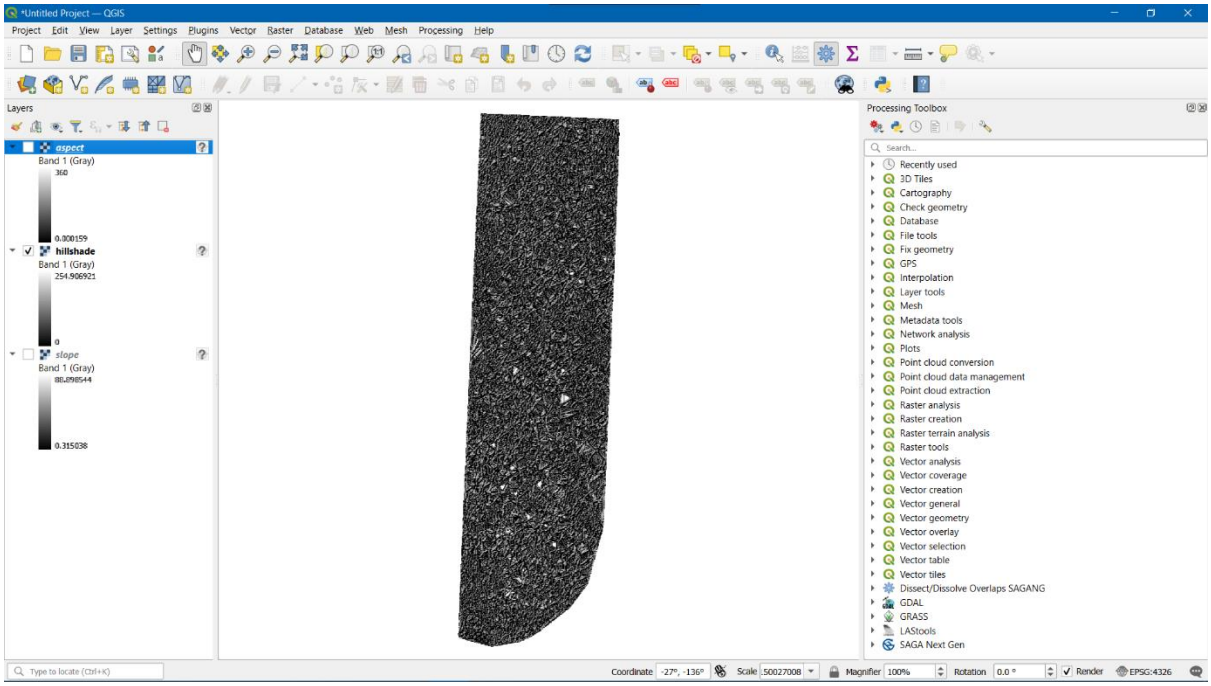
Answer: Generally, **no**, a simple global elevation threshold is not applicable for classifying ground and above-ground points across varying terrain.

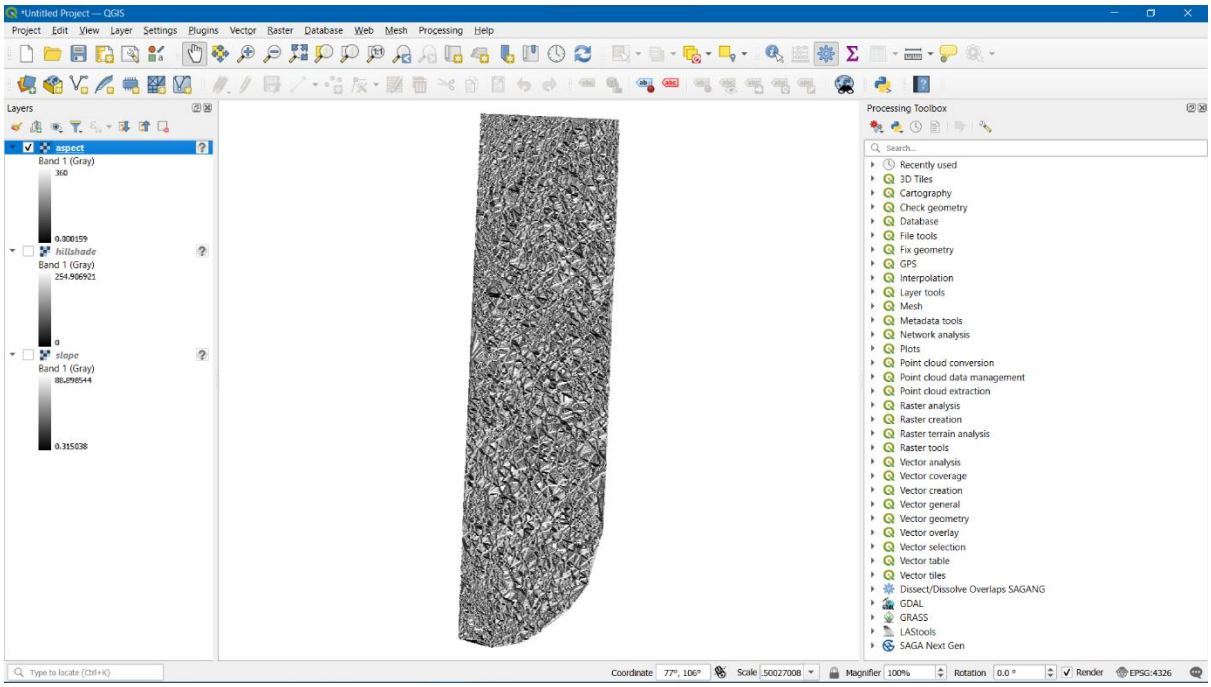
Why: A single elevation threshold assumes the terrain is perfectly flat. In real-world, undulating topography, applying a static elevation threshold fails completely. For example, if you set a threshold based on a valley, all ground points on a nearby hill will be misclassified as "above-ground" objects (like buildings or trees). Conversely, a threshold calibrated for a hill will misclassify low-lying trees in the valley as "ground."

- Generate slope map, Aspect map, shaded relief.
    - Import DEM (.tif) file to Q-GIS (DEM is raster generated for ground class). Make sure your Raster file has spatial Reference. (Coordinate Reference System).
    - Go to ToolBox> Spatial Analyst Tools> Surface> Slope



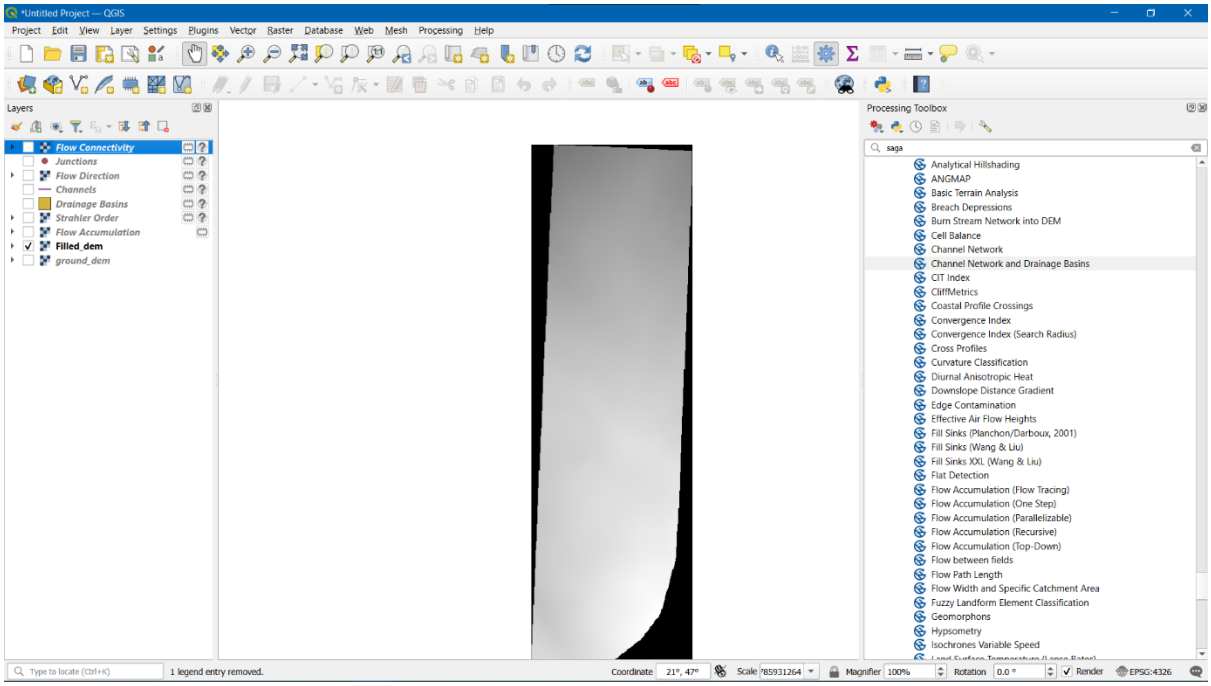
- There are Options to generate Slope, Aspect, Hill shade.
- Select each option. Provide input file as generated Raster. Provide Output path.
- In Method, select Planer. (Later you can try by selecting Geostatic as well.)
- Follow same method for Aspect map and Hillshade.



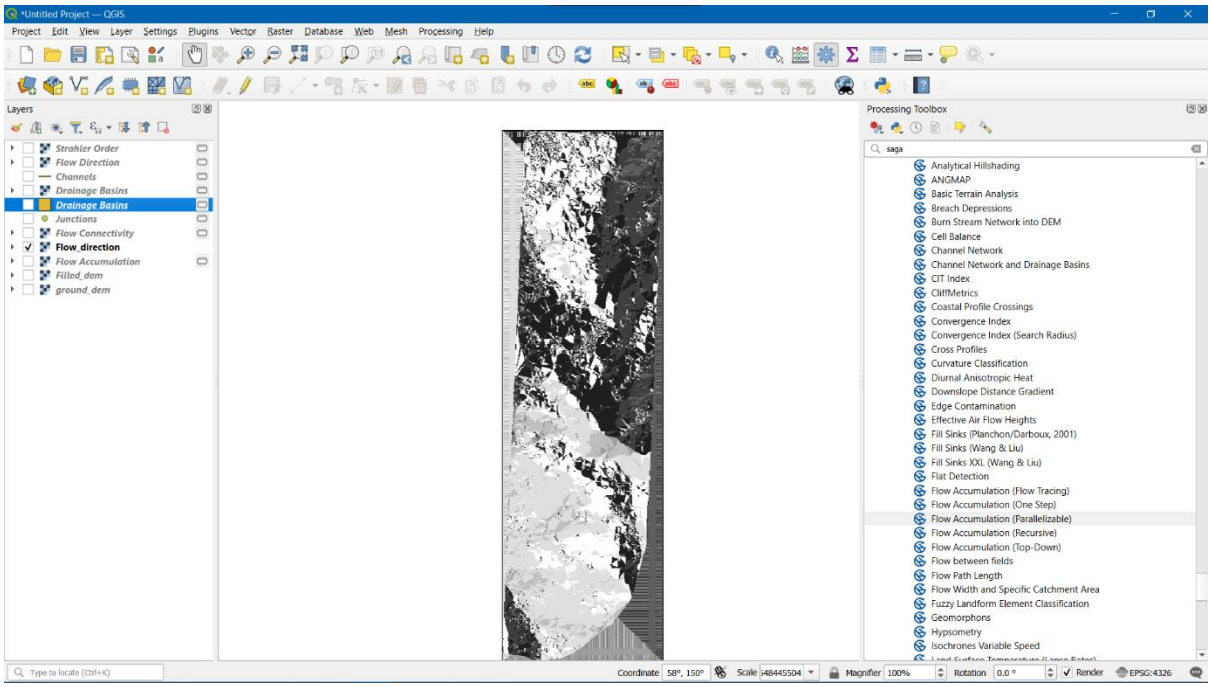


- Generate channel network in the area using Q-GIS:
  - Again, open Raster file in Q-GIS
  - Go to ToolBox> Spatial Analyst Tools> Hydrology> Fill.

Filled DEM (Digital Elevation Model): When you get a raw DEM, it often contains tiny imperfections or artificial "pits" and "sinks"—cells surrounded by higher elevation cells. In reality, water would pool here like a puddle. To model continuous flow across the terrain to the edges of the map, algorithms "fill" these artificial sinks with virtual dirt until water can flow out of them. A Filled DEM is a hydrologically corrected surface ready for analysis.



Flow Direction: Once the DEM is filled, the software looks at every single pixel and asks: *"If a drop of water lands here, which of the 8 surrounding neighbour pixels is the steepest path downhill?"* It assigns a code (often called the D8 algorithm) to represent that direction (North, South, East, West, or diagonals). This creates a map made entirely of tiny arrows pointing downhill.

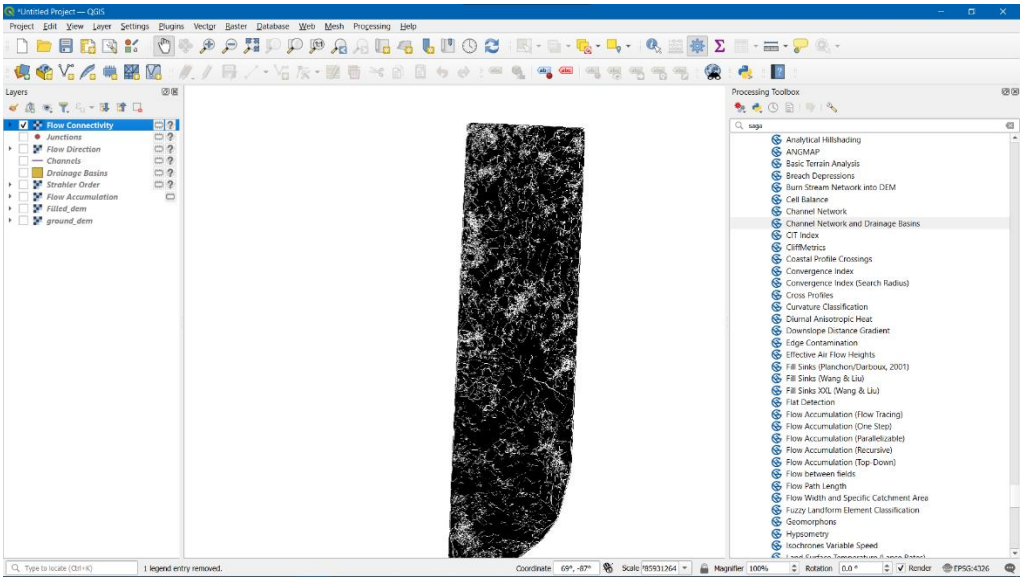


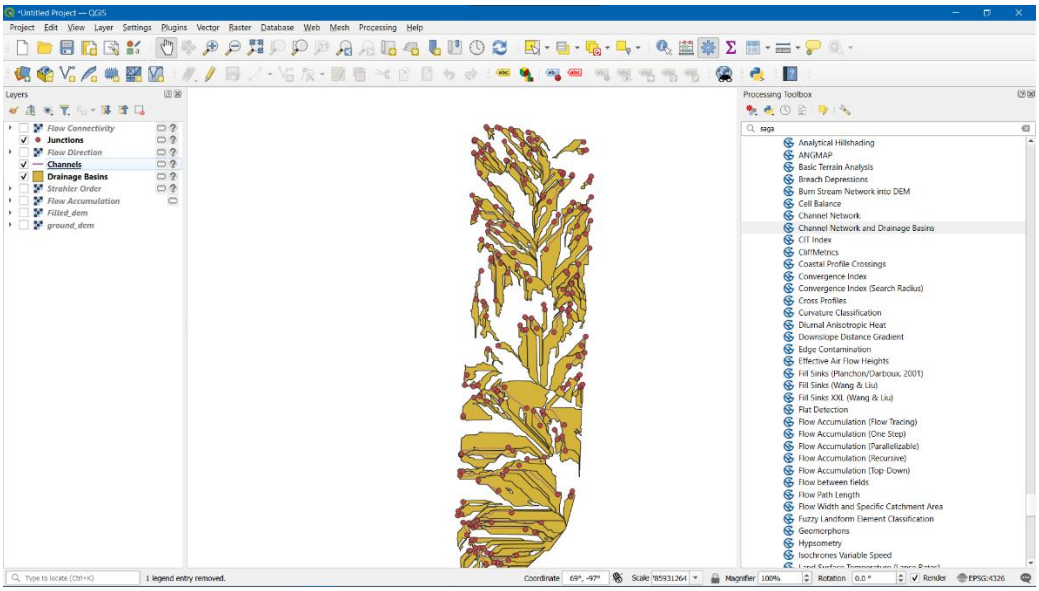
- Select input file and run the process.
- Then, again go to flow ToolBox> Spatial Analyst Tools> Hydrology> Flow direction. Select file which generated after fill tool and get the output.

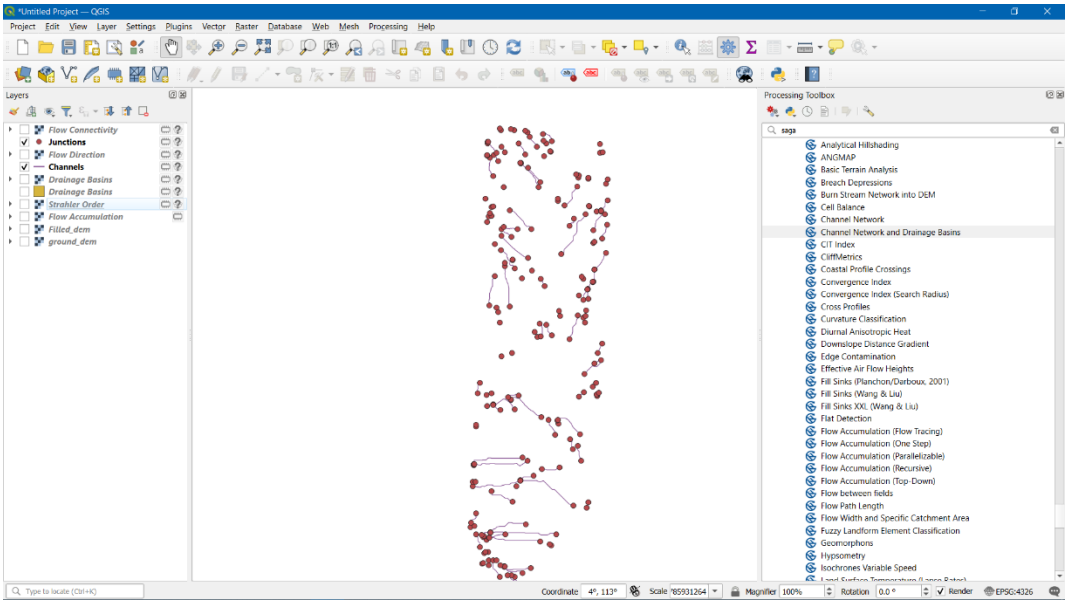
Channel (Stream Network): You cannot map every tiny trickle of water. By applying a "threshold" to the Flow Accumulation raster (e.g., "only show me pixels where more than 300 upstream pixels flow into them"), you define the main rivers and streams. These continuous lines of high flow accumulation are your channels.

Junction (Confluence): These are point features created exactly where two or more distinct channels meet and merge into a single, larger stream.

Flow Connectivity (Stream Links): This refers to the distinct, unbroken segments of the river. A stream link connects a channel's starting point to a junction, or connects two junctions together. This establishes the "routing" so the software knows which river flows into which.



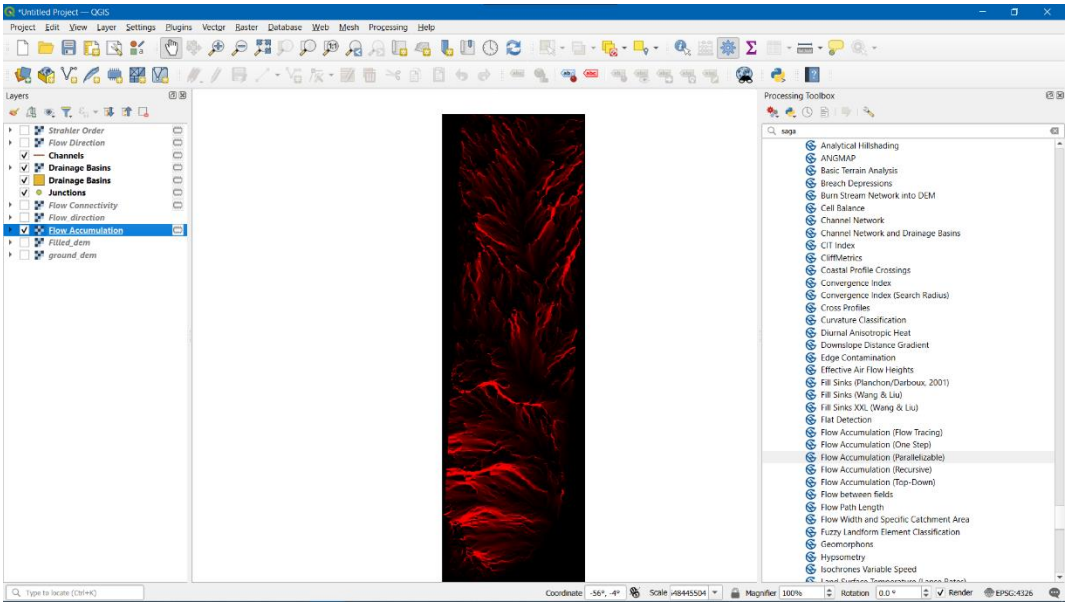




- Again, go to ToolBox> Spatial Analyst Tools> Hydrology> Flow Accumulation.

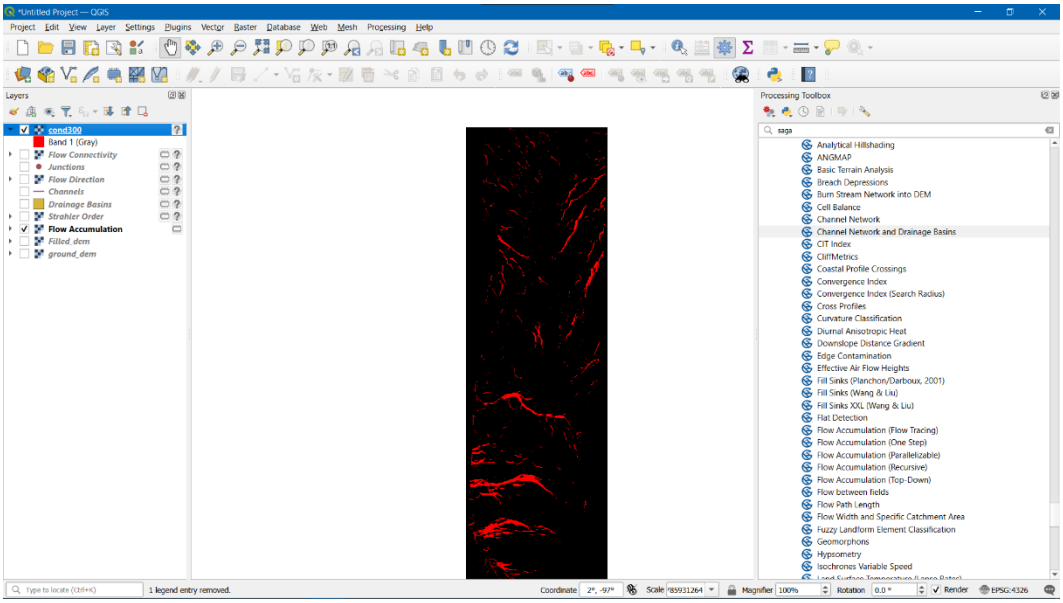
Flow Accumulation: Using the Flow Direction map, the software calculates how many upstream pixels eventually flow into any given pixel.

- Pixels on the top of a hill or ridge will have a value of **0** (no water flows *into* them).
- Pixels at the bottom of a valley will have **huge numbers** (thousands of uphill pixels are contributing water to them).



- For better visualization, go to symbology and try to change color patterns.

- You can also get pixel values which having flow accumulation more than any threshold value. For that, go to ToolBox> Spatial Analyst Tools> Conditional> Con.



- What is the role of threshold for generating Network Analysis.
  Answer: In hydrology modelling within Q-GIS, the Flow Accumulation tool calculates the number of upstream cells that flow into each individual raster cell. The threshold acts as the defining limit that differentiates between overland sheet flow and concentrated channel flow.